Section A

1. SRAM vs DRAM → SRAM uses flip flops to store data, making it faster, expensive and power consuming. It is used in cache memory.
DRAM uses capacitors to store data, needs refreshing, slower but cheaper. It is used as main memory.

2. Programs tend to access a small portion of memory repeatedly.
Type
→ Temporary locality - recently accessed data is reused.
→ spatial locality - nearby locations as accessed.

3. cache memory stores frequently used data/instructions closer to CPU. It reduces avg memory access time and improves performance.

4. Cache Indexing
→ Direct mapped cache
→ set associative cache
→ fully associative cache.

5. Larger cache block size increases spatial locality but increase miss penalty and may load unnecessary data.

6. Associativity is the number of locations where a memory block can be placed in cache. Higher associativity reduces conflict misses but increases hardware cost.

7. Larger cache size reduces miss rate and improves performance but increases cost, power and access latency.

8. i) write through: Data is written to both cache and main memory at the same time.

ii) write-Back: Data is written to cache first and main memory is updated only when the block is replaced.

9. AMAT = Hit time + (miss rate × miss penlty)

time to access cache

Probability of miss

time to fetch block from main memory.

10. LRU block is replaced first. The block which has not been used for the longest time is removed.

11. Memory is divided into mutiple modules and accessed simultaneously in a round robin fashion. It increase memory band width and improves performance.

12. Tracks are concentric circular paths on disk surface. sectors are subdivisions of tracks used to store data blocks.

13. Flash memory is a non-volatile memory that retain data even when power is off. used in pen drives, memory cards.

14. I/O devices have a separate address space. They are accessed using special I/O instructions.

15. I/O devices share the same address space as memory. Devices are accessed using normal memory Instruction.

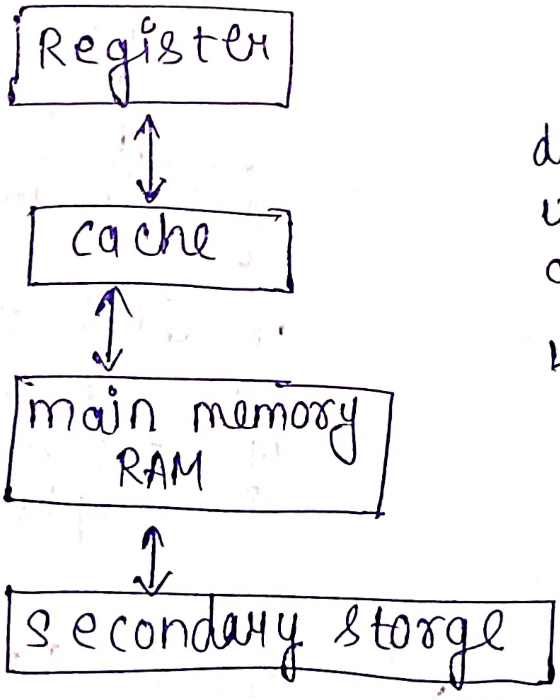
16. CPU contionously checks the status of I/O device and controls the data transfer. CPU is busy waiting.

17. Device interrupts the CPU when it is ready. CPU can perform other tasks meanwhile better CPU utilzation.

18. DMA allows I/O devices to transfer data direectly to/From memory without CPU Involvement. It increase efficiency.

<u>Section = B</u>

1.



Data Flows b/w diff levels.
upper levels are faster and smaller.
Lower Levels are slower and Larger in capacity

2.

Address		
Tag (t bits)	Index (i bits)	offset (b bits)

	Valid	Tag	Data block (2^b bytes)
0			
1			
2			
2^i-1			

Each memory block maps to only one cache line using index.

3. i) write through

→ CPU

↓ write

Cache

↓ write

main memory

each write updates both cache and main memory

ii) write-back

CPU

↓ write

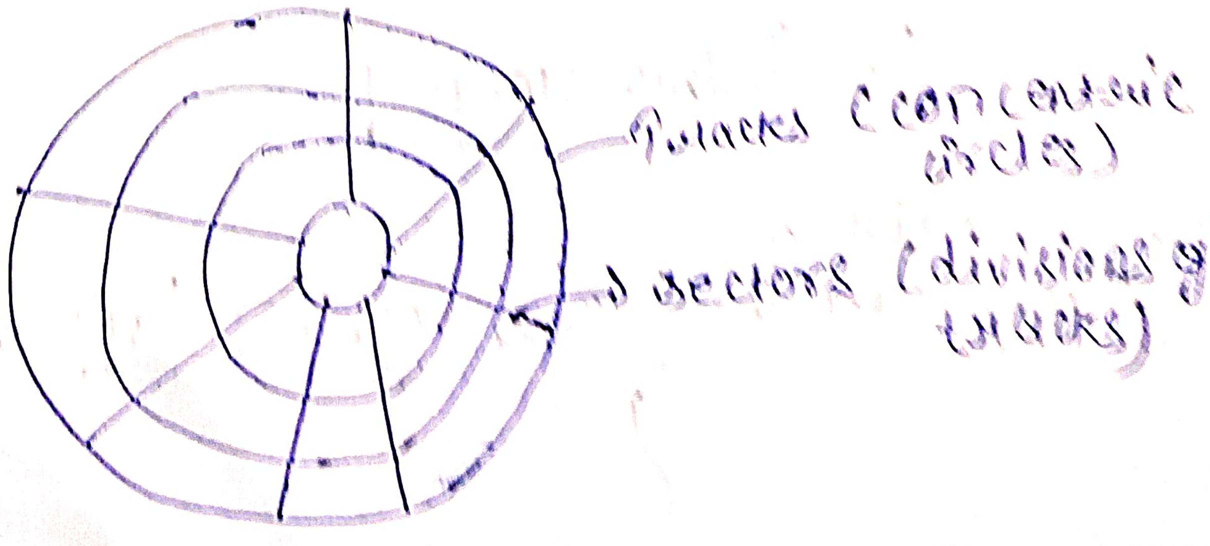
Cache

↕ (write on replacement)

main memory

write updates only cache main memory updated when block is replacement.

4.

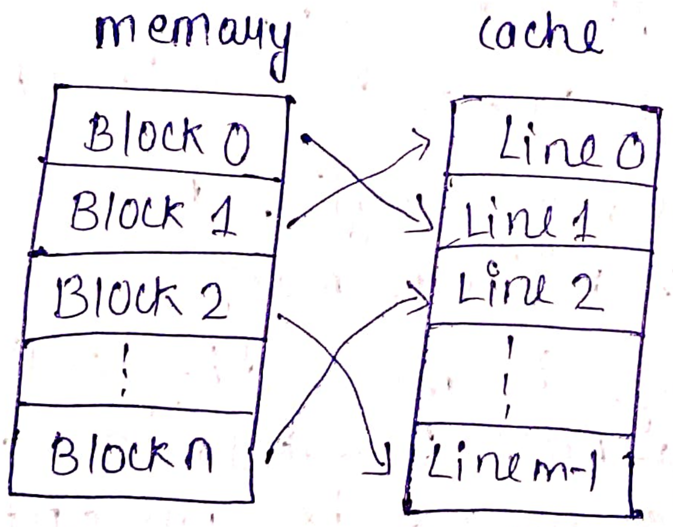


Section - C

1. Temporal locality, because the same location is accessed again and again.
2. Larger blocks increase miss penalty (more data to be transferred) and may load unneccesary data (cache pollution)
3. On every write, data is written to both cache and main memory.
4. Direct memory access (DMA)

Section - D

1.



mapping: memory block is placed in cache line using index. Tag is used in index correctness

2. AMAT = Hit time + (miss rate x miss penalty)

factors affecting performance
- Lower hit time
- Lower miss rate
- Lower miss penalty

These reduces AMAT and improve performance.

3. Programmed I/O	Interupt I/O	DMA
→ CPU checks device continously. → CPU is busy waiting	→ Device interrupt CPU when ready → Better CPU utilization	→ DMA contro-ller transfer data to from memory direcly → Best efficieny

Section - E

1. Two assumption while explaining cache operation.
   - Fixed block size is assumed.
   - Single level cache is considered.
2. Increases memory traffic because every write updates main memory.
3. cache mapping is most difficult because it involves address breakdown and mapping logic which is conceptual and needs practice.